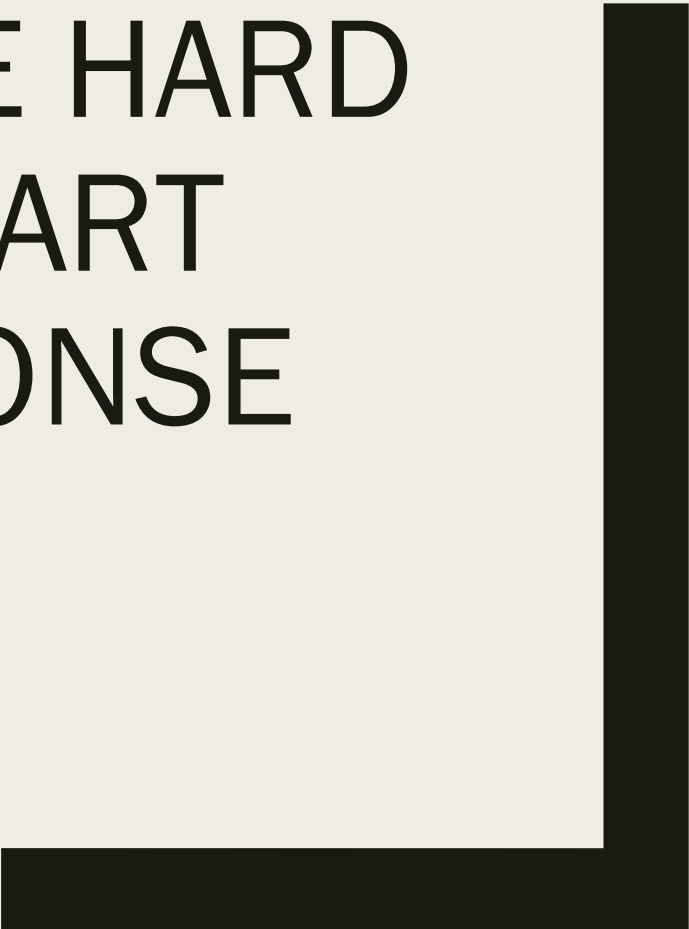




EFFECT OF BASELINE HARD DRUG USE ON HAART TREATMENT RESPONSE

Eduardo Romo
BIOS 6624
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Research Question & Statistical Hypotheses

- Does baseline hard drug use affect treatment response 2 years after HAART initiation?
- Four outcomes:
 - *Mental QoL change*
 - *Physical QoL change*
 - *CD4+ T-cell change*
 - *Log viral load ratio*
- Primary Exposure: baseline hard drug use
- Adjusted for age, BMI, smoking, education, race/ethnicity
- Statistical hypotheses:
 - H_0 : Mean treatment response is equal between groups
 - H_1 : Mean treatment response differs

Data Overview

- Source: Multicenter AIDS Cohort Study
- Final analysis sample: 476 subjs
- Collapsed categorical vars:
 - *Race -> White vs Other*
 - *Smoking -> Current vs Not*
 - *Education -> College vs No college*
- Implausible BMI values set to NA (≤ 0 or ≥ 250)
- 8.2% reported hard drug use
- Hard drug users:
 - *More likely current smokers*
 - *Less likely college educated*

Table 1: Baseline characteristics stratified by baseline hard drug use.

Variable	Overall	No hard drugs	Yes hard drugs
Sample size			
N	476	437	39
BMI			
Mean (SD)	25.2 (4.3)	25.3 (4.4)	23.6 (3.4)
Missing, n (%)	13 (2.7%)	10 (2.3%)	3 (7.7%)
Age			
Mean (SD)	43.3 (8.7)	43.1 (8.7)	44.6 (9.5)
CD4+ Count			
Mean (SD)	373.5 (200.5)	375.4 (201.1)	352.2 (194.7)
Log Viral Load			
Mean (SD)	4.5 (0.9)	4.5 (0.9)	4.5 (0.9)
Mental Quality of Life			
Mean (SD)	44.9 (13.5)	45.1 (13.7)	42.3 (11.2)
Physical Quality of Life			
Mean (SD)	51.0 (9.1)	51.3 (9.1)	47.7 (8.5)
Smoking status			
Not current smoker	291 (61.1%)	282 (64.5%)	9 (23.1%)
Current smoker	185 (38.9%)	155 (35.5%)	30 (76.9%)
Adherence category (year 2)			
$\geq 95\%$ Adherence	426 (89.5%)	388 (88.8%)	38 (97.4%)
$< 95\%$ Adherence	50 (10.5%)	49 (11.2%)	1 (2.6%)
Race/ethnicity			
White	298 (62.6%)	279 (63.8%)	19 (48.7%)
Other	178 (37.4%)	158 (36.2%)	20 (51.3%)
Education			
No College Degree	271 (56.9%)	242 (55.4%)	29 (74.4%)
College Degree	205 (43.1%)	195 (44.6%)	10 (25.6%)

Statistical Analysis

- Models
 - *Linear regression for each outcome*
 - *Bayesian models (primary)*
 - *Frequentist models (validation)*
- Adjustment covariates: baseline outcome value, age, BMI, smoking status, education, and race/ethnicity
- Secondary analysis
 - *Added 2 year adherence to assess mediation between hard drug use and treatment response*
- Inference
 - *Point estimate + 95% intervals*
 - *Bayesian posterior probabilities of clinically meaningful effects*

Main Results: Hard Drug Use & Treatment Response

- Little evidence for mental QoL or viral load effects
- Hard drug use associated with:
 - ↓ *Physical QoL improvement*
 - ↓ *CD4+ recovery*
- Bayesian and frequentist results consistent

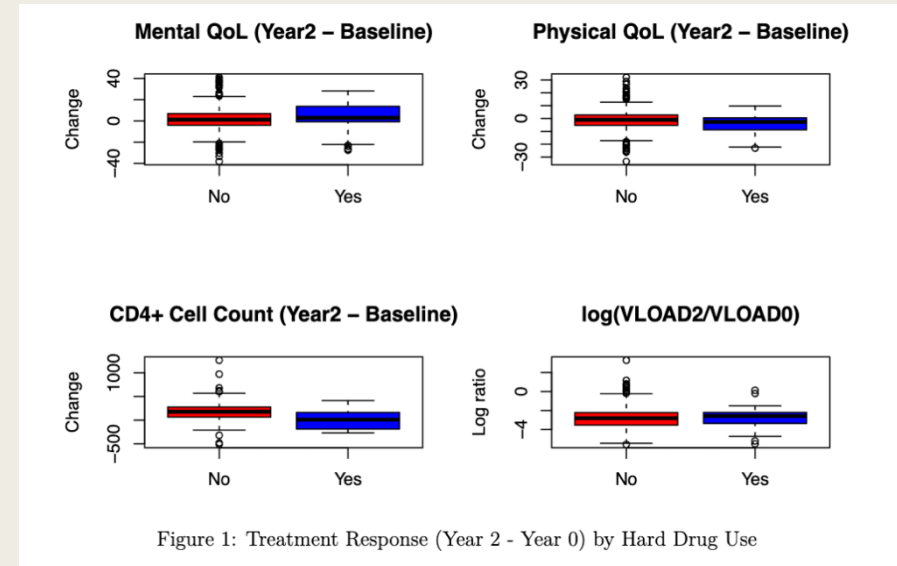


Table 2: Main analysis: hard drug use coefficient by outcome (Frequentist vs Bayesian).

Outcome	Frequentist			Bayesian			
	Estimate (SE)	95% CI	p-value	Post. mean (SD)	95% HPDI	Pr(> clin)	ΔLOOIC (red-full)
Δ AGG_MENT	-0.29 (1.76)	[-3.75, 3.17]	0.8699	-0.29 (1.77)	[-3.78, 3.16]	0.261	-2.9
Δ AGG_PHYS	-3.34 (1.39)	[-6.06, -0.61]	0.0166	-3.34 (1.39)	[-6.05, -0.60]	0.834	3.6
Δ CD4 (LEU3N)	-163.88 (31.79)	[-226.35, -101.41]	<0.001	-148.85 (30.36)	[-209.06, -90.29]	1.000	23.6
log10(VLOAD2/VLOAD0)	-0.06 (0.20)	[-0.46, 0.34]	0.7698	-0.06 (0.20)	[-0.47, 0.33]	0.018	-1.9

Secondary Analysis: Adjustment for Adherence (naïve mediation)

- Goal: assess how hard drug coefficient changes after adjusting for adherence
- Estimates changed minimally across outcomes
- Suggests adherence does not strongly explain observed associations
- Exploratory analysis, not a formal mediation analysis

Table 3: Secondary analysis (naive mediation): hard drug use coefficient by outcome (Frequentist vs Bayesian).

Outcome	Frequentist				Bayesian				
	Estimate (SE)	95% CI	p-value	Δ (with-no)	Post. mean (SD)	95% HPDI	Pr(> clin)	Δ (with-no)	ΔLOOIC (no-with)
Δ AGG_MENT	-0.53 (1.77)	[-4.00, 2.94]	0.7639	-0.24	-0.53 (1.77)	[-3.99, 2.96]	0.279	-0.24	-0.1
Δ AGG_PHYS	-3.57 (1.39)	[-6.30, -0.84]	0.0106	-0.23	-3.57 (1.39)	[-6.30, -0.85]	0.871	-0.23	1.5
Δ CD4 (LEU3N)	-169.68 (31.82)	[-232.21, -107.15]	<0.001	-5.80	-153.79 (30.42)	[-213.89, -94.54]	1.000	-4.93	2.2
log10(VLOAD2/VLOAD0)	-0.01 (0.20)	[-0.41, 0.38]	0.9500	0.05	-0.01 (0.20)	[-0.42, 0.37]	0.014	0.05	3.8

Discussion & Limitations

- Baseline hard drug use associated with:
 - *Worse physical QoL improvement*
 - *Poorer CD4+ recovery*
- Little evidence for mental QoL or viral load effects
- Adherence did not substantially explain associations
- Limitations
 - *Complete case analysis*
 - *Collapsed covariates*
 - *Naïve mediation approach (not causal mediation)*